

Introduction

Objective:

This report details the establishment of a defensible baseline machine learning model for emergency department (ED) triage at Mercer General Hospital. The primary goal is to demonstrate that basic models can outperform random chance in classifying Emergency Severity Index (ESI) levels.

Context:

The project aims to improve patient prioritization in the ED. This phase focuses on building and evaluating two simple classifiers: Logistic Regression and a Decision Tree, to provide a foundation for future development.

Methodology

Dataset:

The project utilizes the Yale EMLC dataset, which contains records of patient visits to the ED, with each row representing a patient encounter and columns representing various clinical and administrative features.

Preprocessing:

Features for model training were carefully selected to exclude leakage variables (outcomes known only after triage), administrative details, and fairness-sensitive demographics. Numerical features were scaled using StandardScaler for the Logistic Regression model. The target variable is esi (Emergency Severity Index).

Model Training:

Logistic Regression: A linear classification model was trained using `LogisticRegression(max_iter=1000, random_state=42)`. This model is valued for its interpretability.

Decision Tree Classifier:

A tree-based model was trained using `DecisionTreeClassifier(max_depth=5, random_state=42)`. The `max_depth` was bounded to 5 to control complexity, prevent overfitting, and maintain interpretability.

Baseline Model:

A `DummyClassifier(strategy="stratified", random_state=42)` was used as a stratified random-guess baseline to provide a minimal performance benchmark against which the trained models could be compared.

Train/Test Split: The dataset was divided into an 80% training set and a 20% test set using `train_test_split`, stratified on the `esi` target variable to ensure proportional representation of each ESI level in both sets. A `random_state=42` was used for reproducibility.

Results and Evaluation

Models Used:

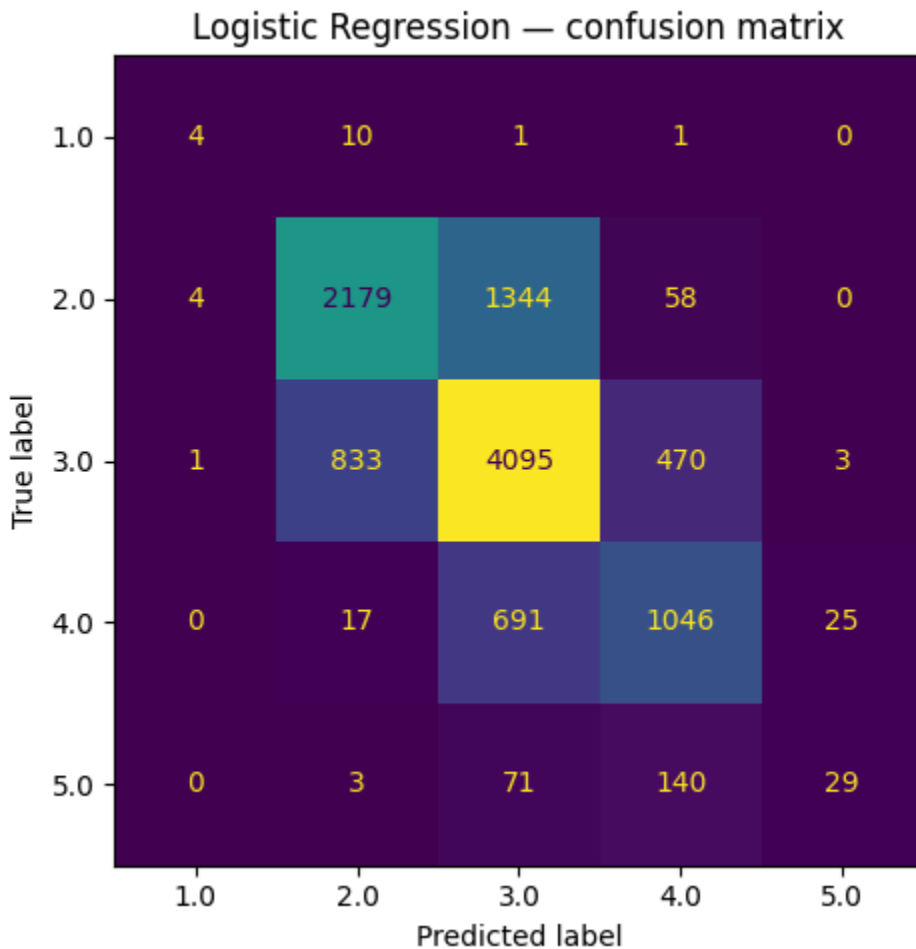
1. Logistic Regression
2. Decision Tree
3. Dummy Baseline

Results Table:

Model	Accuracy	Macro F1	Recall (ESI 1)
Dummy Baseline	0.375	0.204	0.00
Logistic Regression	0.667	0.495	0.25
Decision Tree	0.556	0.216	0.00

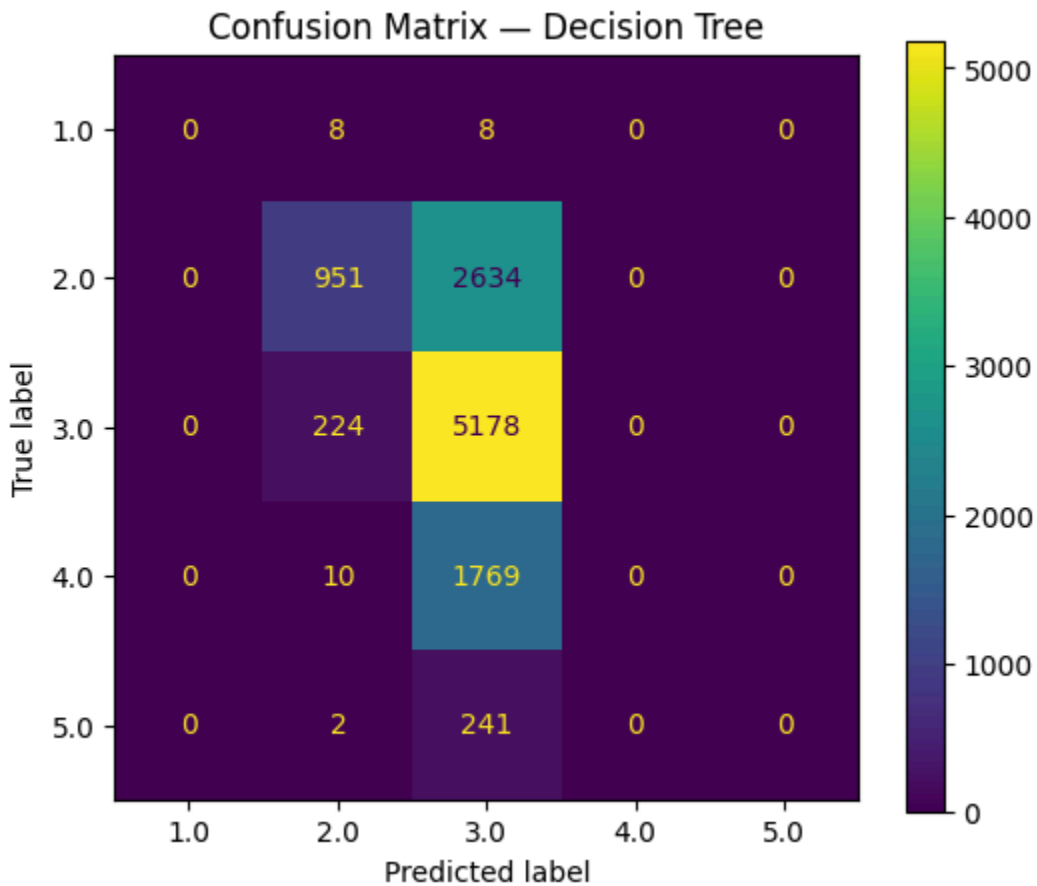
Confusion Matrices:

Figure 1. Logistic Regression Confusion Matrix



The Logistic Regression model correctly classified the majority of ESI Level 2, 3, and 4 patients. However, confusion was observed between neighbouring triage levels, particularly ESI 2 and ESI 3. The model identified some ESI Level 1 patients but still missed a substantial number of critical cases. This highlights the challenge of detecting rare high-acuity patients while maintaining overall accuracy.

Figure 2. Decision Tree Confusion Matrix



The Decision Tree confusion matrix showed poorer performance than Logistic Regression, particularly for ESI Level 1 patients. The model failed to correctly identify any ESI Level 1 cases, producing a recall of 0.00 for the most critical patient group. While the model captured some patterns within the larger ESI classes, its inability to identify the sickest patients makes it less suitable as a clinical baseline.

Classification Report (Logistic Regression):

	Precision	Recall	F1-score	Support
1.0	0.444	0.250	0.320	16
2.0	0.716	0.608	0.658	3585
3.0	0.660	0.758	0.706	5402

4.0	0.610	0.588	0.599	1779
5.0	0.509	0.119	0.193	243
Accuracy			0.667	11025
Macro Avg	0.588	0.465	0.495	11025
Weighted Avg	0.667	0.667	0.661	11025

Primary Metric Justification

Recall for ESI Level 1 was selected as the primary metric because ESI Level 1 patients represent the most critically ill patients in the Emergency Department. Missing a genuine ESI Level 1 patient could delay life-saving treatment. For this reason, reliably identifying critical patients is more important than maximizing overall accuracy.

Macro vs Weighted F1

Macro F1 treats all classes equally, regardless of how many patients belong to each class. Weighted F1 gives greater influence to larger classes. In triage, Macro F1 is useful because rare but clinically important classes such as ESI Level 1 receive equal consideration.

Failure Mode

The most concerning failure mode is misclassifying ESI Level 1 patients as lower-acuity categories. This could delay treatment for critically ill patients and poses greater clinical risk than over-triaging a stable patient. For example, the Logistic Regression model's recall of 0.25 for ESI 1, while better than the Decision Tree, still means a significant proportion of these most critical patients would be missed by the model, posing a significant risk.

Conclusion

Both baseline models outperformed the stratified random baseline. The Logistic Regression model demonstrated an ESI 1 recall of 0.25, while the Decision Tree model achieved 0.00. This indicates that while both models showed potential, the Logistic Regression model offered a superior ability to identify the most critical patients (ESI 1). Recall for ESI Level 1 was identified as the most clinically relevant measure because patient safety depends on reliably

identifying the most urgent cases. These baseline models provide a reasonable starting point for future development, with a clear focus on improving the identification of ESI 1 patients. Future work should focus on improving identification of ESI Level 1 patients through feature engineering, class-balancing techniques, hyperparameter tuning, and more advanced modelling approaches.